

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

INTRODUCTION TO THE HUMAN BODY

Levels of Structural Organization

(Smallest to Largest)

1. **Chemical Level** - most basic level of organization
 - Consists of elements, atoms, and molecules such as proteins, carbohydrates, inorganic compounds and essential biochemicals
 - Foundation of all living things
 - Building blocks for all structures and functions in the body
 - **Atoms** - smallest unit of an element composed of protons, neutrons, and electrons
 - **Molecules** - formed when 2 or more atoms bond together (Ex: Water, Glucose)
 - **Macromolecules** - large and complex molecules made from smaller units (Ex: Proteins, Carbohydrates, Lipids, Nucleic Acids)
 - **Organic Compounds** - carbon-based molecules essential for life
 - **Inorganic Compounds** - molecules that typically do not contain carbon (Ex: Water, Salts, Acids, Bases)
 - **Chemical Bonds** - forces that hold atoms together in molecules (Ex: Ionic Bonds, Covalent Bonds)
2. **Cellular Level** - where individual cells are the basic structural and functional units of life
 - Includes all life processes that occur within a single cell
3. **Tissue Level** - group of similar cells that work together to perform a specific function, along with their surrounding intercellular material
 - A. **Epithelial Tissue** - covers the body's surfaces and lines its cavities and organs
 - B. **Connective Tissue** - provide support and connection for other tissue
 - C. **Muscle Tissue** - specialized for contraction and movement
 - D. **Nervous Tissue** - made of neurons and glial cells, responsible for transmitting electrical signals to control and coordinate the body's functions
4. **Organ Level** - structure of 2 or more tissue types working together for specific functions
5. **Organism/Organismal Level** - represents the culmination of all previous levels, forming a complete human being capable of independent function

Importance:

- *Without structural organization, humans could not survive or function, because life depends on the organized interaction of all biological levels*

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

Basic Life Processes

1. **Metabolism** - sum of all chemical reactions (both building up and breaking down) that occur in the body to maintain life
 - *Ex: When you eat, your body digests and breaks them down into nutrients like glucose and amino acids, which are then converted into energy.*
2. **Responsiveness** - ability to detect and respond to changes in the internal or external environment
 - *Ex: On a hot day, your body detects this temperature and responds by sweating to cool itself down.*
3. **Movement** - includes motion of the entire body, individual organs, single cells, and even structures within cells
 - *Ex: A ballerina performing a pirouette is an example of large body movement as it uses many muscles and joints to turn the whole body.*
4. **Growth** - increase in body size due to an increased number of cells, the size of existing cells, or the amount of material around cells
 - *Ex: A teenager developing bigger muscles because of proper nutrition and exercise shows how the body increases in size through cell growth.*

5. **Differentiation** - developmental process where unspecialized cells change into specialized cells with distinct structure and function
 - *Ex: Stem cells in the bone marrow develop into red blood cells that carry oxygen or white blood cells that fight infections, showing how cells become specialized.*
6. **Reproduction** - biological process where living organisms produce new individuals or offspring, or the process of the formation of new cells
 - *Ex: A married couple having a baby, ensuring the continuation of the human species.*

Homeostasis

- Existence and maintenance of a relatively constant environment within the body
 - Internal equilibrium of the body as changes in external environmental conditions can result in changes in internal body conditions
 - Network of organ systems that helps keep the body's internal environment relatively constant
1. **Negative Feedback Mechanism** - when any deviation from the set point is made smaller or is resisted
 - more commonly involved in maintenance of homeostasis
 - *Ex: Maintenance of body temperature*

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

2. **Positive Feedback Mechanism** - occur when response to the original stimulus results in the deviation from the setpoint becoming even greater
 - *Ex: Blood loss being resolved by blood clotting ; Giving birth*

Organ Systems

1. **Integumentary System** - provides protection, regulates temperature, prevents water loss, and helps produce vitamin D
 - **Components**: Skin, Hair, Nails, Sweat glands, Sebaceous glands
2. **Skeletal System** - Provides framework, support, protection, movement, mineral storage, and blood cell formation.
 - **Components**: Bones, Cartilages, Ligaments, Joints
3. **Muscular System** - Produces movement, maintains posture, and generates heat.
 - **Components**: Muscles (Skeletal, Smooth, and Cardiac)
4. **Nervous System** - Controls and coordinates body activities, processes information, and maintains homeostasis
 - **Components**: Brain, Spinal cord, Nerves, and Sensory receptors

5. **Endocrine System** - Regulates long-term processes (growth, reproduction, metabolism) through hormones
 - **Components**: Hypothalamus, Pituitary gland, Pineal gland, Thyroid gland, Parathyroid glands, Thymus gland, Adrenal glands, Pancreas, Ovaries, Testes
6. **Cardiovascular System** - Transports oxygen, nutrients, hormones, and wastes; circulates blood
 - **Components**: Heart, Arteries, Veins, Capillaries, Blood
7. **Lymphatic System** - Removes foreign substances from the blood and lymph, combats disease, maintains tissue fluid balance, and absorbs dietary fats from the digestive tract
 - **Components**: Lymph nodes, Lymphatic vessels, Spleen, Thymus, Tonsils
8. **Respiratory System** - Exchanges oxygen and carbon dioxide between the blood and air and regulates blood pH
 - **Components**: Nasal cavity, Pharynx, Larynx, Trachea, Bronchi, Bronchioles, Alveoli, Lungs

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

9. **Digestive System** - Performs the mechanical and chemical processes of digestion, absorption of nutrients, and elimination of waste
 - **Components:** Oral cavity, Pharynx, Esophagus, Stomach, Small intestine, Large intestine, Liver, Gallbladder, Pancreas
10. **Urinary System** - Removes waste products from the blood and regulates blood pH, ion balance, and water balance
 - Components: Kidneys, Ureters, Bladder, Urethra
11. **Reproductive System (Male and Female)** - Produces gametes and hormones, supports reproduction
 - **Components (Male):** Testes, Epididymis, Vas deferens, Prostate gland, Penis
 - **Components (Female):** Ovaries, Fallopian tubes, Uterus, Vagina, Mammary glands

Body Planes and Regions

- **Plane** - imaginary flat surfaces passing through the body
 - divides the body, making it possible to “look inside” and observe the body’s structures
 - **Section** - a cut made from a plane
1. **Sagittal Plane** - separates the body or a structure into right and left halves
 - refers to the way the body would be split by an arrow passing anteriorly to posteriorly

2. **Frontal (Coronal) Plane** - divide the body into front (anterior) and back (posterior) halves
3. **Transverse (Horizontal) Plane** - runs parallel to the ground, dividing the body into superior and inferior portions
4. **Oblique Plane** - a plane angled other than horizontal or vertical
 - is not parallel or at a right angle

Directional Terms

- **Anatomical position** - refers to a person standing with the face directed forward, the upper limb hanging to the sides, and the palms of the hands facing forward

Term	Etymology'	Definition	Example
Right		Toward the right side of the body	Right ear
Left		Toward the left side of the body	Left eye
Superior	L. higher	A structure above another	The chin is superior to the navel.
Inferior	L. lower	A structure below another	The navel is inferior to the chin.
Cephalic	G. <i>kephale</i> , head	Closer to the head than another structure (usually synonymous with <i>superior</i>)	The chin is cephalic to the navel.
Caudal	L. <i>cauda</i> , a tail	Closer to the tail than another structure (usually synonymous with <i>inferior</i>)	The navel is caudal to the chin.
Anterior	L. before	The front of the body	The navel is anterior to the spine.
Posterior	L. <i>posterus</i> , following	The back of the body	The spine is posterior to the breastbone.
Ventral	L. <i>venter</i> , belly	Toward the belly (synonymous with <i>anterior</i>)	The navel is ventral to the spine.
Dorsal	L. <i>dorsum</i> , back	Toward the back (synonymous with <i>posterior</i>)	The spine is dorsal to the breastbone.
Proximal	L. <i>proximus</i> , nearest	Closer to the point of attachment to the body than another structure	The elbow is proximal to the wrist.
Distal	L. <i>di-</i> plus <i>sto</i> , to stand apart or be distant	Farther from the point of attachment to the body than another structure	The wrist is distal to the elbow.
Lateral	L. <i>latus</i> , side	Away from the midline of the body	The nipple is lateral to the breastbone.
Medial	L. <i>medialis</i> , middle	Toward the midline of the body	The nose is medial to the eye.
Superficial	L. <i>superficialis</i> , toward the surface	Toward or on the surface (not shown in figure 110)	The skin is superficial to muscle.
Deep	O.E. <i>deap</i> , deep	Away from the surface, internal (not shown in figure 110)	The lungs are deep to the ribs.

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

PHYSICAL THERAPY SETTINGS

1. Acute Care Hospital Setting

- **Definition:** refers to hospitals where patients receive short-term, immediate medical and rehabilitative care for severe injuries, illnesses, or pre-surgery (2-4 weeks) or post surgery cases
- focuses on stabilizing patients, preventing complications, and initiating early mobility
- recovering from trauma, acute illness, or surgery
- **Common Complications:**
 - Pneumonia
 - Pressure Ulcers
 - Deep Vein Thrombosis (DVT)
- **Interventions:**
 - Bed mobility training
 - Breathing exercises and airway clearance techniques
 - GAIT Training (Early ambulation and mobility exercises)
 - Positioning to prevent pressure sores and contractures
 - Education on safety and body mechanics
 - Patient transfers (latter part)
- **Patients/Conditions Involved:**
 - Post-surgical patients (e.g., after orthopedic, cardiac, or abdominal surgery)
 - Stroke or traumatic brain injury patients
 - Burn injury patients
 - Accident or trauma victims

2. Inpatient Rehabilitation Setting

- **Definition:** provides intensive rehabilitation services to patients who remain in the facility (admitted to hospital) for several days or weeks.
- seen 3x a day for 2-3 hours
- Interdisciplinary team approach by collaboration of OTs, SLPs, Physicians, Nurses
- **Interventions:**
 - Strength and endurance training
 - Balance and coordination training
 - Functional mobility and transfer training
 - Neuromuscular re-education
 - Activities of Daily Living (ADL) retraining / Task oriented training
- **Patients/Conditions Involved:**
 - Stroke, spinal cord injury (SCI), and traumatic brain injury (TBI) patients
 - Orthopedic rehabilitation (e.g., joint replacements, fractures)
 - Post-amputation rehabilitation / Post-surgery
 - Severe trauma recovery

3. Outpatient / Ambulation Care Setting

- **Definition:** involves patients who live at home but visit clinics or rehabilitation centers for scheduled PT sessions (come in and go)
- care is less intensive
- patients are medically stable

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

- **Interventions:**

- Therapeutic exercises for strength, flexibility, and endurance
- Manual therapy and soft tissue mobilization
- Modalities (ultrasound, electrical stimulation, heat/ice therapy)
- Postural correction and ergonomic training

- **Patients/Conditions Involved:**

- Musculoskeletal injuries (sprains, strains, back/neck pain, LBP)
- Post-operative patients (e.g., ACL repair, rotator cuff surgery)
- Sports-related injuries (ACL tear, Patellar dislocation)
- Chronic pain and orthopedic disorders
- Neurological conditions (e.g. mild stroke)

4. **Skilled Nursing Facility (SNF)**

- **Definition:** provides rehabilitation care for elderly patients

- less intensity and more of a homecare treatment

- **Interventions:**

- Balance and coordination training
- Endurance training
- ADL training
- Fall prevention strategies
- Functionality training
- Education on assistive device use

- **Patients/Conditions Involved:**

- Elderly patients with age-related mobility issues
- Chronic conditions (e.g., Parkinson's, arthritis)
- Cardiopulmonary patients

5. **School-Based Setting**

- **Definition:** focuses on providing physical therapy services within educational institutions to support students with physical disabilities or developmental delays

- ensures that patients can participate effectively in school activities

- **Interventions:**

- Gross motor skill development
- Balance and coordination exercises
- Position/postural control and body awareness training
- Adaptive equipment training (wheelchairs, braces, walkers)
- Collaboration with teachers and parents for accessibility support
- Parent & patient education

- **Patients/Conditions Involved:**

- Children with cerebral palsy, muscular dystrophy, or spina bifida
- Developmental delay
- Speech delay
- Autism spectrum disorder
- Orthopedic or neurological impairments affecting school participation

INTRODUCTION TO PT AND MED TERMS

MIDTERMS | 1st SEMESTER

6. Wellness, Sports, and Fitness Setting

- **Definition:** emphasizes injury prevention, athletic performance enhancement, and overall health promotion
- **Interventions:**
 - Strength and resistance training
 - Plyometric exercises
 - Injury prevention and biomechanical assessment
 - Sports-specific training and return-to-play programs
 - Posture correction and core stability exercises
 - Education on healthy movement patterns
- **Patients/Conditions Involved:**
 - Athletes recovering from sports injuries (e.g., ACL tear, sprain, tendinitis)
 - Individuals aiming for weight management or improved physical performance
 - People engaged in fitness programs to prevent injury

7. Community-Based Rehabilitation

- **Definition:** involves bringing rehabilitation and wellness services directly to local communities
- often focuses on health promotion, disability prevention, and improving quality of life
- **Interventions:**
 - Community exercise (e.g. zumba, cardio, flexibility)
 - Seminars on healthy lifestyles and ergonomics
 - Collaboration with local health organizations

● Patients/Conditions Involved:

- Stroke survivors
- Elderly individuals at risk of falls
- Persons with disabilities or chronic illnesses
- Populations in underserved or rural areas

8. Homecare Setting

- **Definition:** to provide rehabilitation at the comfort of the patient's own home environment
- **Interventions:**
 - Home exercise programs tailored to the environment
 - Functional mobility and ADL training
 - Family/caregiver education
 - Pain management and strengthening exercises
- **Patients/Conditions Involved:**
 - Post-surgical or post-hospitalization patients (hip fracture, knee replacement)
 - Geriatric patients
 - Neurological patients (e.g., stroke, Parkinson's, paralyzed)